

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

Slot A

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5) **Assessment Tool Weightage: 40%**

QUESTIONS:5 Total Marks 50

Duration :60 Minutes Annexure A

Industry Problem Solving Task

Code of Conduct:

I _____M.THANUJA(192425356)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.THANUJA

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	

Professionalism	5%	Highly professional, welldocumented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	
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Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation Models available:

Viola-Jones: 78%, low latency

YOLO: 92%, moderate latency

Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

ANS. Smart Surveillance System Selection

Given constraints:

- Latency > 100 ms → system fails
- Accuracy < 85% → unacceptable
- Hardware supports only moderate computation

Model	Accuracy	Latency	Computation
Viola-Jones	78%	Low	Low
YOLO	92%	Moderate	Moderate
Deep Learning	95%	High	High

Analysis:

Viola-Jones has low latency and low computational requirements, but its accuracy is only **78%**, which is below the required **85%**. Therefore, it cannot be selected.

Deep Learning provides the highest accuracy of **95%**, but it has high latency and requires high computation. Since the hardware supports only moderate computation and latency above 100 ms causes failure, it may not satisfy the deployment requirements.

YOLO provides **92% accuracy**, which is above the required 85%, and has moderate latency and moderate computational requirements.

Decision:

YOLO is the best choice because it satisfies the accuracy requirement while matching the moderate computational capability of the hardware.

Conclusion:

YOLO provides the best balance between **accuracy, latency, and computational requirements** and is therefore suitable for the smart surveillance system.

2. Autonomous Vehicle

Detection Logic A vehicle

detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

ANS. Autonomous Vehicle Detection Logic

Requirements:

- Accuracy > 90%
- Response time < 50 ms

Method Accuracy Response Time Satisfies Requirements?

A	88%	30 ms	NO
B	93%	70 ms	NO
C	91%	45 ms	YES

Analysis:

Method A has a fast response time of 30 ms, but its accuracy is only 88%, which is below the required 90%. Therefore, it does not satisfy the safety requirement.

Method B has 93% accuracy, which is acceptable, but its response time is 70 ms. Since the required response time is less than 50 ms, Method B is unsuitable.

Method C has **91% accuracy** and a response time of **45 ms**. Therefore, it satisfies both requirements.

Decision:

Method C should be selected.

Conclusion:

Method C is the only option satisfying both the **accuracy > 90%** and **response time < 50 ms** constraints. Since autonomous vehicles require both safety and fast response, Method C is the most suitable choice.

3. Motion Analysis Decision

Problem A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

ANS. Motion Analysis Decision Problem

Requirements:

- Accurate motion-direction detection
- Robustness in dense traffic

Approach	Advantage	Limitation
Optical Flow	Accurate motion information	Noisy in dense scenes
Motion Estimation	Stable	Less detailed

Analysis:

Optical Flow can provide accurate information about the direction and movement of objects. However, in dense traffic, multiple vehicles move close to each other. This can introduce noise and make the motion vectors unreliable.

Motion Estimation provides more stable results and is therefore more robust in dense traffic. However, it provides less detailed motion information compared with Optical Flow.

Since the problem specifically requires **robustness in dense traffic**, stability is important.

Decision:

Motion Estimation is more suitable for high-density traffic conditions.

Although Optical Flow is more accurate in ideal conditions, its sensitivity to noise in dense scenes makes it less reliable for this particular scenario.

Conclusion:

For a traffic system operating under high-density conditions, **Motion Estimation should be preferred because its stability provides better robustness**, even though some motion detail may be lost.

4. Background Modeling Evaluation A GMM-based system produces:

- **Static scenes: 90% accuracy**
- **Dynamic scenes: 65% accuracy**

Requirement:

System must maintain $\geq 80\%$ accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

ANS. Background Modeling Evaluation

Given GMM performance:

Scene	Accuracy
Static	90%
Dynamic	65%

Requirement: Accuracy must be $\geq 80\%$ in all scenarios.

Analysis:

In static scenes, GMM achieves **90% accuracy**, which satisfies the requirement of at least 80%.

However, in dynamic scenes, GMM achieves only **65% accuracy**. This is 15 percentage points below the required 80%.

Therefore, GMM performs well in static environments but poorly in dynamic environments.

Does GMM alone satisfy the requirement?

No.

Because the system must maintain at least 80% accuracy **across all scenarios**, the 65% accuracy in dynamic scenes causes GMM to fail the requirement.

Suggested Decision

GMM should **not be used alone**. A more robust approach should be considered for dynamic environments, such as combining background modeling with another detection or tracking technique.

Conclusion:

GMM is suitable for static scenes but is **insufficient as a standalone solution** because its dynamic-scene accuracy is only 65%.

5. Depth Estimation Strategy

Constraints:

- **Only one camera available**
- **Moderate depth accuracy acceptable**

Stereo Vision: high accuracy, needs two cameras

Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

ANS. Depth Estimation Strategy

Constraints:

- Only one camera is available.
- **Moderate depth accuracy** is acceptable.

Method	Accuracy	Camera Requirement
Stereo Vision	High	Two cameras
Structure from Motion	Moderate	One camera

Analysis:

Stereo Vision provides high depth accuracy by using images from two cameras. However, the given system has only one camera. Therefore, Stereo Vision cannot satisfy the hardware constraint.

Structure from Motion (SfM) can estimate depth using a **single camera** and provides moderate depth accuracy. Since moderate accuracy is acceptable, it satisfies both requirements.

Decision:

Structure from Motion should be selected.

Conclusion

Structure from Motion is the most feasible solution because it:

- Requires only **one camera**
- Provides **moderate depth accuracy**
- Satisfies the given hardware and accuracy constraints

Therefore, **SfM is more suitable than Stereo Vision for this application**